

Generic Hybrid PV-BESS Model in an Electromagnetic Transients Simulator for Transmission Connected Plants

User Manual for PSCAD™ Model Prototype



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User Manual for PSCAD™ Model Prototype

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Software Manual, March 2025

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SOFTWARE DESCRIPTION

This manual documents the implementation of a generic model of a hybrid PV-BESS plant suitable to represent transmission-connected plant in the electromagnetic transient (EMT) simulator PSCAD™.

Description

The hybrid PV-BESS model is formed by three individual modules, the power plant controller, the PV inverter and the BESS inverter module. The PV inverter model contains the necessary control loops to meet the fault ride-through performance requirements specified in IEEE Std 2800-2022. The BESS inverter model provides flexibility to select various control modes including grid following and grid forming modes. This manual includes an example parameterization of the model and presents example simulation results that demonstrate the fault ride-through performance of the hybrid PV-BESS plant.

Benefits and Value

The model could be used by transmission planners in long-term planning studies as well as to assess potential system impacts of a prospective hybrid PV-BESS plant at an early stage in its interconnection process when a manufacturer-specific inverter model may not yet be available.

Platform Requirements

This model is compatible with 64-bit Windows 10, and Windows 11. Third-party EMT domain (PSCAD™ v5 or higher) software is required.

Keywords

Generic EMT Model, IEEE 2800-2022, Solar PV, BESS, Model Prototype.

ACRONYMS

BESS: Battery energy storage system

EMT: electromagnetic transients

FRT: fault ride-through

IBR: inverter-based resource

PI: proportional plus integral (controller)

PLL: phase locked loop

POI: Point of interconnection

PPC: power plant controller

PPS: primary power source

PV: photovoltaic

SCR: short circuit ratio

SMIB: Single machine infinite bus

CONTENTS

1 INTRODUCTION1

2 MODEL SETUP AND SPECIFICATIONS2

Hybrid plant module2

BESS plant module5

PV plant module8

SMIB module10

3 EXAMPLE RESULTS11

Case 1: Unbalance fault11

Case 2: Frequency step change13

Case 3: Voltage step change.....16

Case 3a: BESS in Grid following mode16

Case 3b: BESS Grid forming mode18

4 CONCLUSION.....20

5 REFERENCES21

LIST OF FIGURES

Figure 1. EPRI PV BESS Hybrid Model Main page	2
Figure 2. EPRI Hybrid PV BESS plant module	3
Figure 3. BESS inverter model	5
Figure 4. The average inverter circuit model	6
Figure 5. BESS inverter parameters	6
Figure 6. BESS inverter parameters	7
Figure 7. PV inverter model	8
Figure 8. PV inverter model parameters	9
Figure 9. PV inverter model parameters	10
Figure 10. Case 1: Fundamental frequency components of the PV inverter terminal voltage and current	12
Figure 11. Case 1: Active power, reactive power and RMS voltage at the PV inverter terminal.....	12
Figure 12. Case 1: Fundamental frequency components of the BESS inverter terminal voltage and current	13
Figure 13. Case 1: RMS voltage, active power and reactive power at the POI	13
Figure 14. Case 2: Active power, reactive power, voltage and currents at the POI	14
Figure 15. Case 2: Voltages, active power, reactive power and currents at the HV side of the BESS terminal.	15
Figure 16. Case 2: Total active power, reactive power and voltage from the PV	15
Figure 17. Case 3a: RMS voltage, active power and reactive power at the POI	16
Figure 18. Case 3a: Fundamental frequency components of the BESS inverter terminal voltage and current	17
Figure 19. Case 3a: Fundamental frequency components of the PV inverter terminal voltage and current	17
Figure 20. Case 3b: RMS voltage, active power and reactive power at the POI	18
Figure 21. Case 3b: Fundamental frequency components of the BESS inverter terminal voltage and current	19
Figure 22. Case 3b: Fundamental frequency components of the PV inverter terminal voltage and current	19

LIST OF TABLES

Table 1. Plant-level voltage control4

Table 2. Plant level active power control4

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1 INTRODUCTION

This report documents a generic hybrid PV-BESS three phase inverter model developed by EPRI in the EMT simulation PSCAD software format. The use of the particular simulation software is by choice and the specifications listed in this document can be equally adopted into any particular EMT software.

The hybrid PV-BESS plant model combines a solar and a battery energy storage plant model that share the same point of interconnection (POI) and uses a power plant controller to control the output of the two plants as an inverter-based resource (IBR). The development of this hybrid model relies on the sources given in Section 5. The model is generic and does not represent any specific equipment manufacturer.

The following model limitations should be considered when using this hybrid model:

1. The example control structure and algorithm demonstrated for the generic hybrid plant is not the only way to ensure stable operation. There are other methods presently available in the industry and the one chosen for this model is neither an endorsement nor a suggestion that other methods were considered inferior.
2. The values of controller gains and other settings used in the model are generic and may not be suitable or appropriate for:
 - a. Different ratings or operating conditions
 - b. Different network topologies
 - c. Different load dynamic characteristics
 - d. Different number of varied source device characteristics

However, for the above conditions, it is possible to tune the controller gains and other settings to achieve satisfactory response of the model.

3. The design and control of hybrid plant varies from manufacturer to manufacturer. As a result, it is not expected for the generic hybrid plant model to provide exact match to the response of commercial plants, particularly during transient period. However, the model is expected to provide the general trend of response.
4. The model is not expected to predict controller instability resulting from a specific design or control implementation. As such, the model prototype should not be used for the analysis of sub-synchronous resonance between PV plants and grid elements, nor for the studying of super-synchronous control interactions between various PV plants.
5. The model represents only one version of a possible hybrid control plant hierarchy. There are numerous other options available and the individual components within the model can be changed around to suit a desired configuration.

2 MODEL SETUP AND SPECIFICATIONS

This chapter describes the setup of the hybrid PV-BESS inverter EMT model, lists its parameters.

The PSCAD model consists of two modules at the main page: a hybrid PV-BESS plant module connected to a single machine infinite bus system (SMIB) module, as shown in Figure 1. User can specify the following parameters that are available on the main page:

Plant_Rating – The total hybrid plant rating in MVA (PV+BESS).

Contract_MW – The total contracted MW of the hybrid plant.

Pref_PV – The required MW output from the PV plant.

Pref_BESS – The required MW output from the BESS.

Vref – The voltage setpoint of the hybrid plant in per unit.

Sys_Freq – The system frequency in Hz.

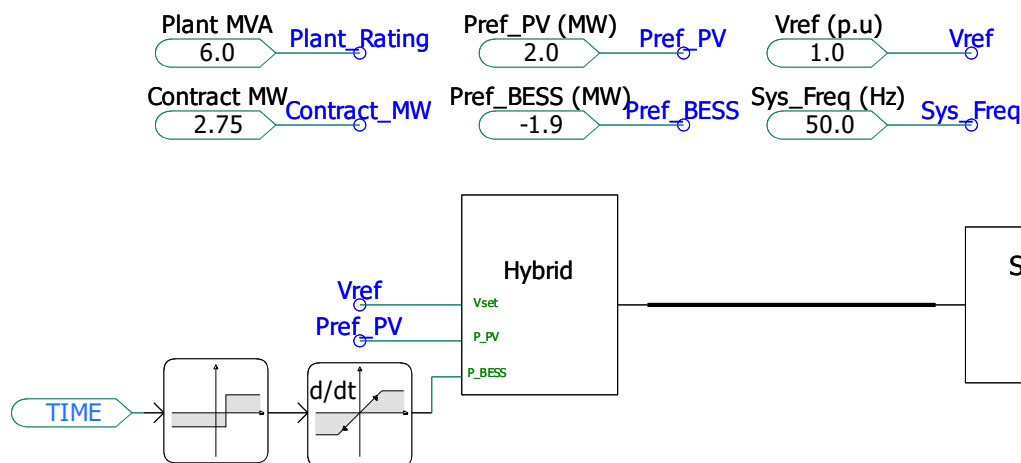


Figure 1. EPRI PV BESS Hybrid Model Main page

Hybrid plant module

The hybrid PV-BESS plant module consists of the following submodules:

- power plant controller (PPC)
- PV model and
- BESS model

The PV plant model consists of an aggregated PV inverter module connected to the POI via a 0.48/12.47 kV step up transformer and a medium voltage line. The PV inverter module consists of a detailed representation of the DC power input from the primary power source module (PPS), an average inverter model and a LC filter model.

The BESS module consists of an aggregated (average) inverter model with constant DC voltage source, a LC filter models, a 0.65/12.47 kV step up transformer and a medium voltage line. The output of the BESS module is connected to the same POI as the PV plant model.

The PPC determines the errors of the active power and voltage measured at the POI calculated against their setpoints and add them to the active power and voltage/reactive power setpoints of the PV and BESS model accordingly.

The following parameters are accessible by the user at the main page of the hybrid PV-BESS plant module.

Parameters for BESS plant:

- BESS_UN – Number of BESS inverter
- BESS_MVA – The total MVA rating of the BESS plant
note: This should be consistent with the product of the number of BESS inverter*Rated MW (specified in the BESS submodule)

Parameters for PV plant:

- UN – Number of PV inverter
- Inv_PV_MVA – Rating of individual PV inverter
- Del_Qref_PV – Reactive power output setpoint of the PV plant

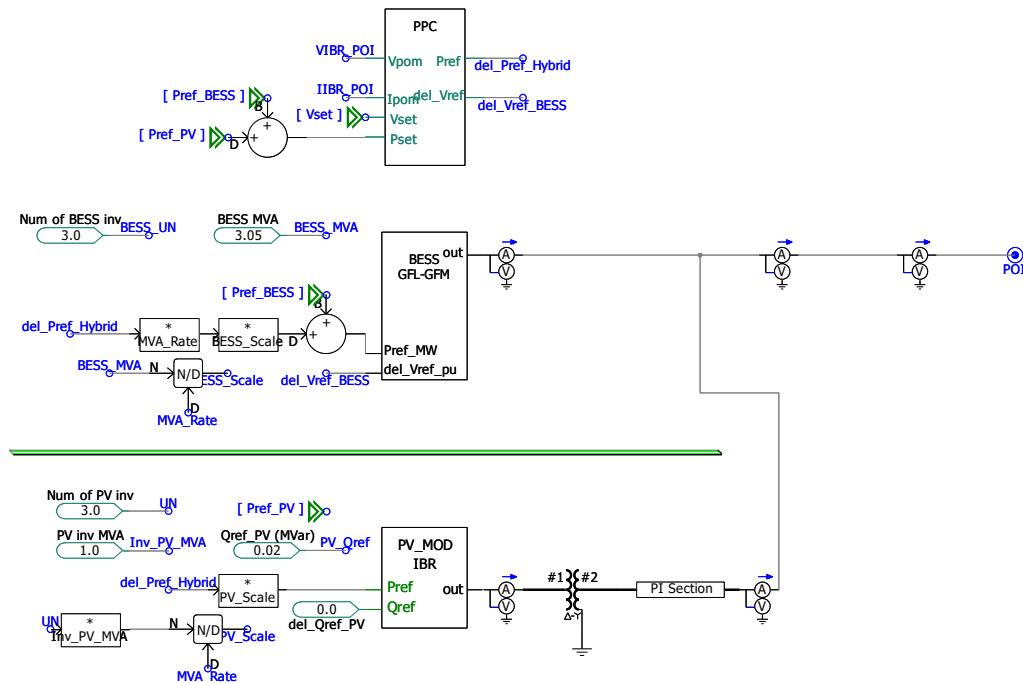


Figure 2. EPRI Hybrid PV BESS plant module

The power plant controller (PPC) measures the instantaneous voltage and currents at the POI and computes the active and reactive power output of the hybrid plant. The PPC module contains a plant level voltage control and a plant level active power control. The function of the plant level active power control loop is to calculate the error between the active power setpoint (taking into consideration the frequency droop setting) and the measured active power output. The calculated error value (del_Pref_Hybrid) is then added proportionally to the active power setpoints (Pref) of the PV and BESS plant.

The PPC module also includes a QV control loop where its output is used to adjust the reference voltage setpoint which is used by the BESS plant to control voltage at the POI. Note that the PV plant model is set up with fixed Q mode.

User can specify the voltage base of the point of measurement at the top level of the PPC component (right click | Edit Parameters). In addition, user can also specify the following parameters in the PPC module:

Table 1. Plant-level voltage control

Parameter	Value
Tc	0.01 [s]
Kc	0 [pu]
dbd	0.02 [pu]
Kp	0.01 [pu/pu]
Ki	1 [pu/pu]
qvrmax	99 [pu]
qvrmin	99 [pu]
Tdelay_out	0.1 [s]
V_LVRT	0.9 [pu]
V_HVRT	1.1 [pu]
Tfltr	0.01 [s]

Table 2. Plant level active power control

Parameter	Value
dprmax	99 [pu]
dprmin	99 [pu]
Tdelay_in	0.01 [s]
pimax	1 [pu]
pimin	-1.0 [pu]
Kpg	0.1 [pu]

Parameter	Value
Kig	1.0 [pu]
Prmax	1.0 [pu]
Prmin	-1.0 [pu]
PPC_enable	1.0 [pu]
Freq_ref	1 [pu]
Tfrq	0.05 [s]
fdbd1	-0.0005 [pu]
fdbd2	0.0005 [pu]
Ddn	20 [pu]
Dup	20 [pu]

Detailed descriptions of the PPC module are presented in [1].

BESS plant module

The BESS inverter model is shown in Figure 3. The following parameters can be specified on the main page of the BESS module.

- Rated_MW – Rated MW of individual inverter
- Vref – Voltage setpoint
- Vbase – Rated MV voltage

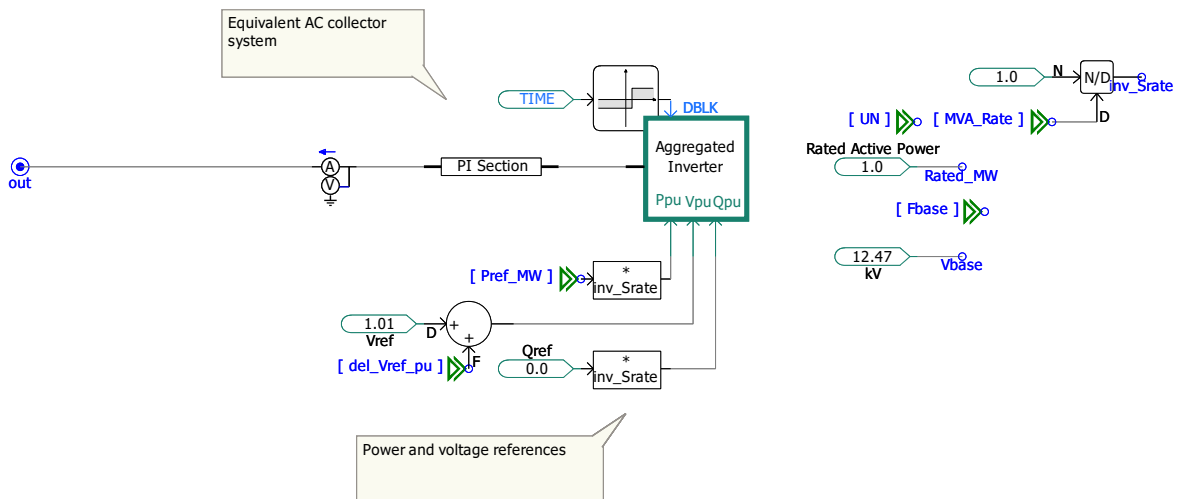


Figure 3. BESS inverter model

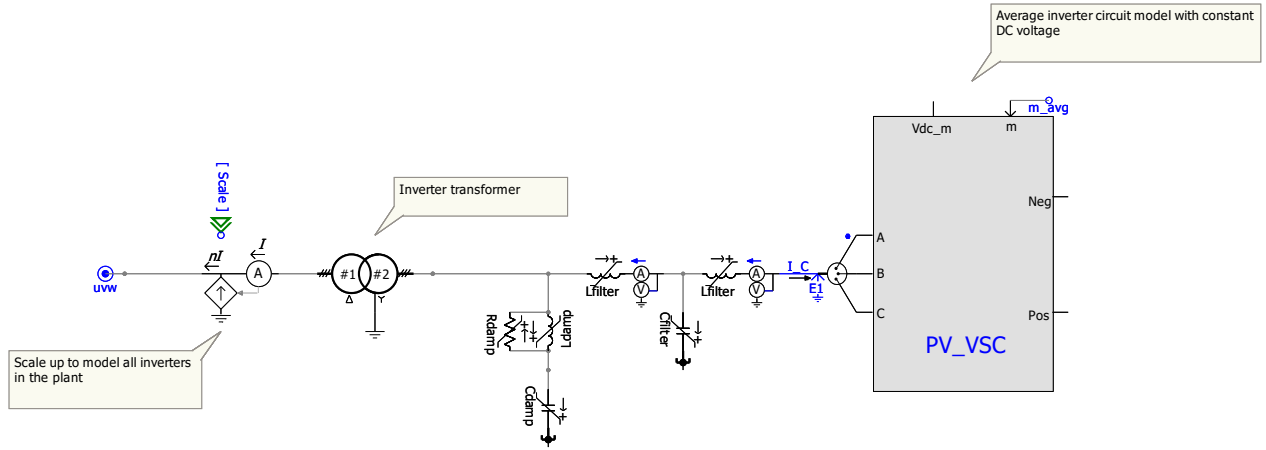


Figure 4. The average inverter circuit model

The average inverter circuit model with constant DC voltage is shown in Figure 4. The inverter model can be set up to be operating in grid following or grid forming mode. Detailed descriptions of the inverter model are presented in [2] and [3]. Figure 5 and Figure 6 show parameters available for user to modify in accordance with the BESS inverter specifications.

▼ Base Values		
Number of Inverters		UN
System frequency (Hz)		Fbase
Base power of the inverter (MVA)		Rated_MW
Base medium voltage (kV, L- to- L)		Vbase
Base inverter voltage (kV, L- to- L)		0.65
Base DC voltage [kV]		1.4
Base leakage inductance of the inverter [pu]		0.2
▼ Limits		
Priority		Q
P max		1.0 [pu]
P min		-1.0 [pu]
Q max		1.0 [pu]
Q min		-1.0
Limit proportional gain		5.0
Limit integral gain		0.1

Figure 5. BESS inverter parameters

		
▼ Active power control (PLL based)		
Active Power Proportional Gain		0.5
Active Power Integral Gain		20
▼ Current control		
D-axis Current Proportional Gain		0.5
D-axis Current Integral Gain		20.0
Q-axis Current Proportional Gain		0.5
Q-axis Current Integral Gain		20.0
▼ Dc voltage control		
MPPT (0); P_control (1)		1
Dc Voltage Proportional Gain		1
Dc Voltage Integral Gain		20.0
▼ General		
Q Flag (0-Q, 1-V)		1
V Flag (0-V, 1-Q/V)		1
Maximum current magnitude		1.3
Low voltage state freeze threshold		0.8
High voltage state freeze threshold		1.3
Time of freeze current after voltage recovery		0.01
▼ Negative sequence control		
Neg seq control flag (Kfactor - 0; Vneg elimination - 1)		0
Neg seq current limit		1.3
Kfactor control gain		5
Kfactor control phase angle (degrees)		-90
Vneg elimination control proportional gain		5
Vneg elimination control integral gain		100
▼ Reactive power control (PLL based)		
Reactive Power Proportional Gain		0.5
Reactive Power Integral Gain		20.0
Dynamic voltage support (DVS) deadband		0.1
Dynamic voltage support (DVS) gain		1
▼ Synchronization		
W evaluation type		SRF-PLL
PLL Proportional Gain		20.0
PLL Integral Gain		200.0
Frequency Droop Gain		20.0
Voltage Droop Gain		20.0
RoCoF Gain		0
RoCoF Time Constant		0.01
VSM Damping		0.3
VSM Mechanical Time constant		0.01
▼ Voltage control (non PLL based)		
Voltage (non-PLL) Proportional Gain		3
Voltage (non-PLL) Integral Gain		10
▼ Voltage control (PLL based)		
Voltage (PLL) Proportional Gain		0.5
Voltage (PLL) Integral Gain		100.0

Figure 6. BESS inverter parameters

PV plant module

The average PV inverter model is shown in Figure 7. Detailed descriptions of the PV inverter model are presented in [1]. Figure 8 and Figure 9 show parameters available for user to modify in accordance with the PV inverter specifications. Note that the model is set up such that the upper limit for the Pref signal from the PPC is set to 0.0, thereby preventing the PV from providing frequency response, i.e only BESS will response to the frequency control action from the PPC.

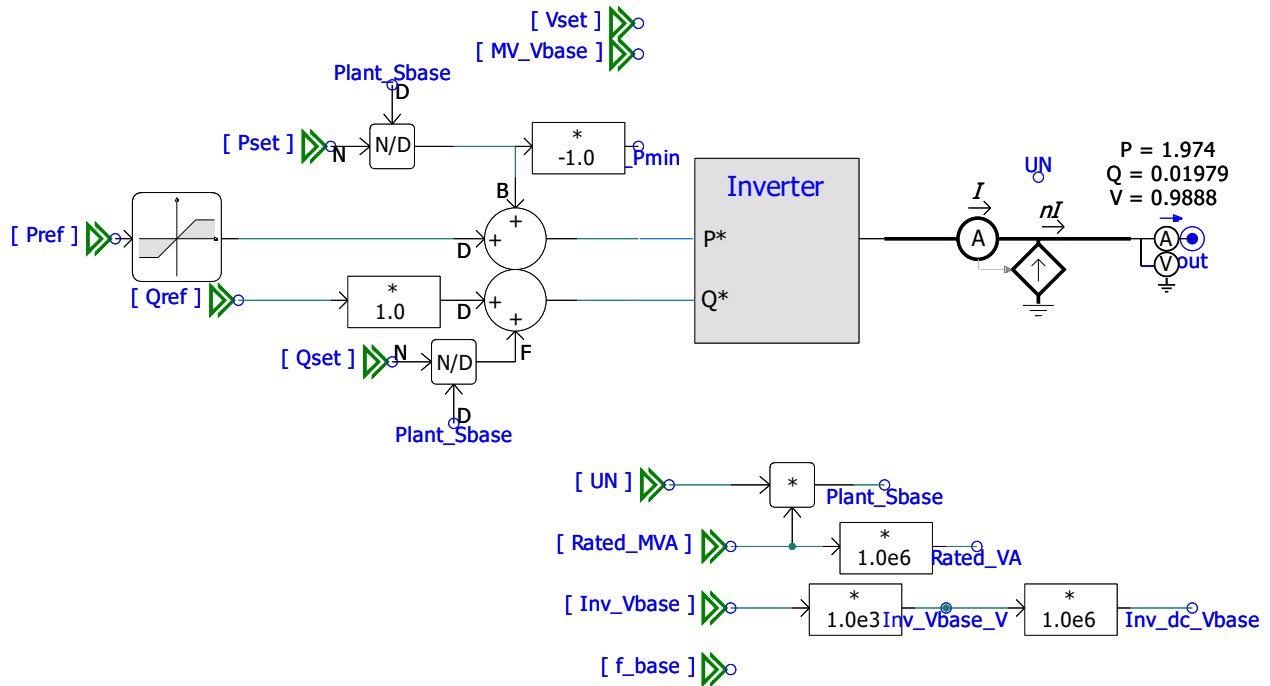


Figure 7. PV inverter model

control configuration	
Vsp_flg (1: enable inverter term. voltage control, 0: Q control)	0
Qsp_CL_flg (1: closed loop Q control, 0: open loop Q control)	1
qt_flg (1: Q control at terminal after LC filter, 0: Q control before LC filter)	1
Vdc_flg (1: enable Vdc control, 0: voltage source on DC bus)	1
PV_flg (1: enable PV array I-V characteristic, 0: constant I)	0
MPPT_flg (1: enable MPPT for Vdc*, requires VI_flg=1 and Vdc_flg=1)	0
f_flg (1: to enable frequency-droop control, 0: disable)	0
current control	
kp_cur: Proportional gain current controller	0.9
ki_cur: Integral gain current controller	900.0
limcurmax: anti-windup upper limit	99999
limcurmin: anti-windup lower limit	-99999
Vff_flg: (1: enable voltage feedforward, 0: disable)	0
Tvff: Vff LPF time constant	0.0001
ip_ramp_up: Ramp rate for increasing active current (pu/sec)	1.0
DC Link	
Vdc_nom: DC link voltage nominal	Inv_dc_Vbase
Cdc: DC link capacitor capacitance (uF)	0.1e6
frequency-droop	
fdbd1: deadband for frequency droop control (< 0.0)	-0.0006
fdbd2: deadband for frequency droop control (> 0.0)	0.0006
Dup: 1/Droop for low frequency	20.0
Ddn: 1/Droop for high frequency	20.0
Tfrq: 1st order LPF time constant for PLL frequency	0.1
FRT	
Ilim_pu: current limit	1.2
Kqv1: FRT V1 proportional gain	1.0
Vhvrt: threshold for HVRT	1.1
VLvrt: threshold for LVRT	0.9
Kqv 2: FRT V2 proportional gain	0
V2_db: V2 FRT dead band	0.1
ipri_flg: current priority (1: iq, 0: ip)	1
DelI_flg	1
General	
Sinv: Inverter rated apparent power (VA)	Rated_VA
VLL_base: inverter rated RMS voltage line-line (V)	Inv_Vbase_V
fnom: nominal frequency	f_base
tstart_up: Start-up time	0.1
LC filter	
Lf: LC filter inductance (pu)	0.104719
Rf: LC filter resistance (pu)	0.0020833
Cf: LC filter capacitance (pu)	0.01995
Rd: LC filter damping resistance (pu)	0.13888

Figure 8. PV inverter model parameters

▼ PLL		
kp_pll: Proportional gain for PLL		40
ki_pll: Integral gain for PLL		4
limpllmax: PLL integrator windup limit (rad/s)		70
▼ PWM		
fs: PWM switching frequency (switching model)		3060.0
▼ reactive power - voltage control		
Qmax: reactive power maximum limit		0.4
Qmin: reactive power minimum limit		-0.4
iq1_max: reactive current maximum limit		99
iq1_min: reactive current minimum limit		-99
Kqp: Q closed-loop proportional gain		0.5
Kqi: Q closed-loop integral gain		40
Kvp: terminal voltage control proportional gain		0.5
Kvi: terminal voltage control integral gain		100
vt_ref: terminal voltage control reference setpoint		1.00
Tv: Voltage LPF time constant		0.01
▼ signal processing		
Ifilt_flg: Current filter flag (1: enable)		1
Tfilt_i: Current LPF time constant		0.00001
Vfilt_flg: Voltage filter flag (1: enable)		1
Tfilt_v: Voltage LPF time constant		0.00001
▼ Vdc control		
kp_vdc: Vdc proportional gain		5.18
ki_vdc: Vdc integral gain		52.91

Figure 9. PV inverter model parameters

SMIB module

The SMIB module contains an infinite bus test system that can be set up to represent the required equivalent network source based on the hybrid PV-BESS plant rating under test.

The following pre-defined simulation tests can be activated in the SMIB module:

- Voltage changes from 1.0 pu to 0.975 pu and back to 1.0 pu.
- Frequency changes from 50 Hz to 47 Hz and back to 50 Hz.
- Fault at the POI.

3 EXAMPLE RESULTS

This chapter documents and discusses simulation results obtained from running various example cases using the hybrid PV-BESS model.

Case 1: Unbalance fault

Simulation results for a B-C fault at the POI are presented here with the model configuration as documented in the section **Error! Reference source not found..** The BESS plant is represented in grid following mode, which controls P and Q/V with some amount of frequency support at the inverter level.

The PV plant is operating at fixed P and Q control mode. At the inverter level, the positive sequence voltage control during the FRT is enabled while the negative sequence voltage control during the FRT is disabled ($K_{qv2} = 0$).

The fault is applied at $t=10s$ which lasted for 200 ms.

The response of the fundamental frequency positive and negative sequence components of the PV inverter terminal voltage during the fault are shown in Figure 10. The plot shows the inverter responses to the positive sequence voltage drop by injecting reactive current in the positive sequence and since the negative sequence voltage control is disabled, the reactive current in the negative sequence is regulated to zero during the fault. The active current remains constant throughout the fault duration. Figure 11 shows the active power, reactive power output from the PV plant and the RMS voltage at the POI.

The positive and negative sequence voltages of the BESS inverter terminal are shown in Figure 12. The results show that the BESS inverter responded to the positive and negative sequence voltage deviation by changing the positive and negative sequence reactive currents accordingly.

The overall hybrid PV-BESS plant active power, reactive power and the RMS voltage at the POI are shown in Figure 13.

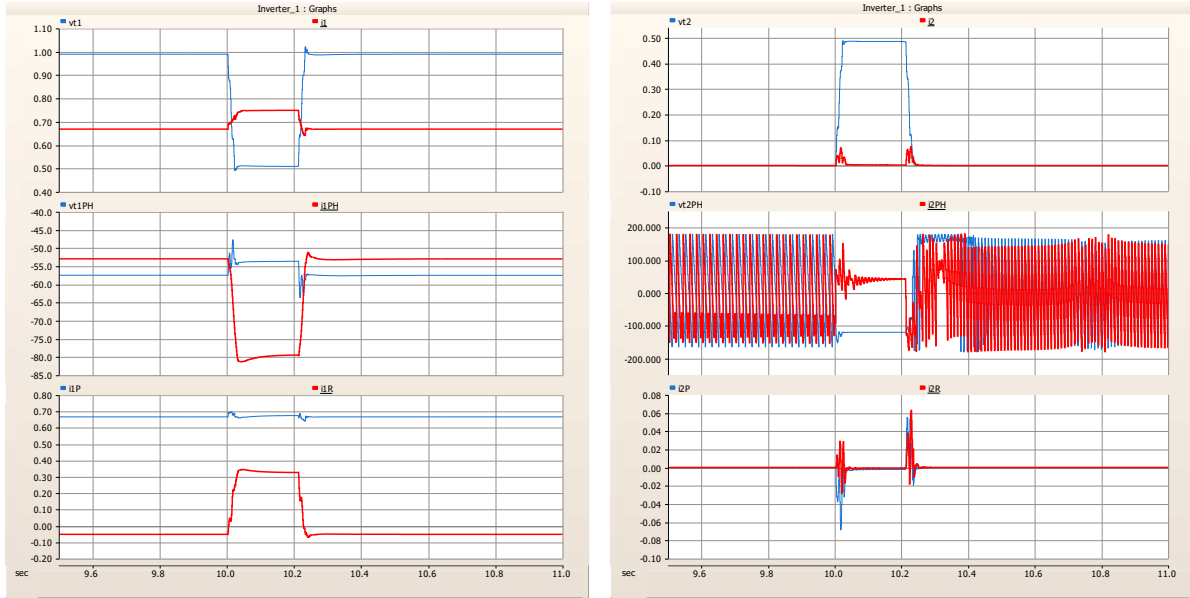


Figure 10. Case 1: Fundamental frequency components of the PV inverter terminal voltage and current

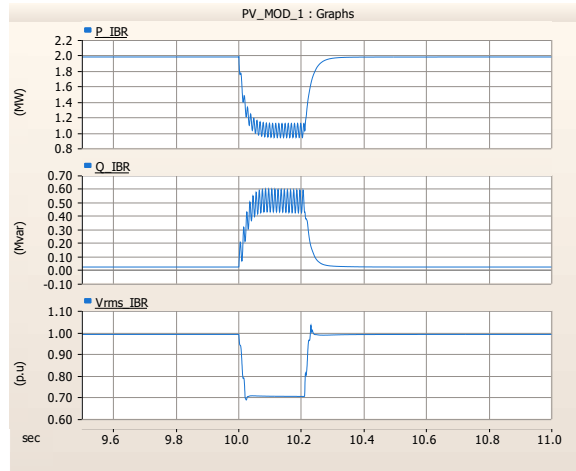


Figure 11. Case 1: Active power, reactive power and RMS voltage at the PV inverter terminal

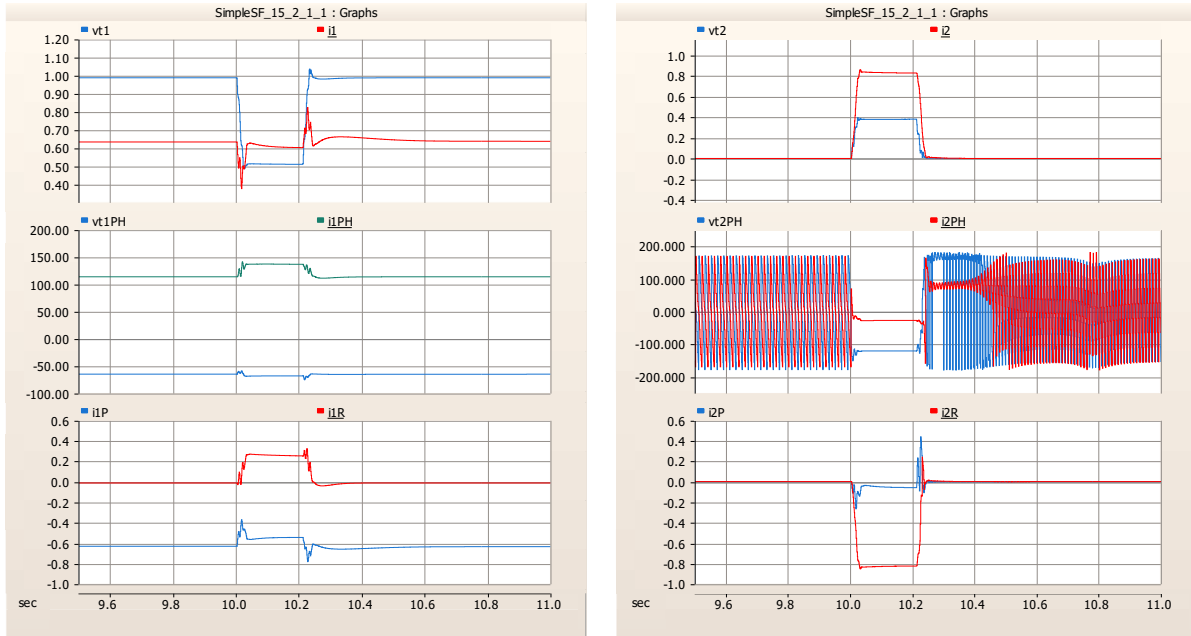


Figure 12. Case 1: Fundamental frequency components of the BESS inverter terminal voltage and current

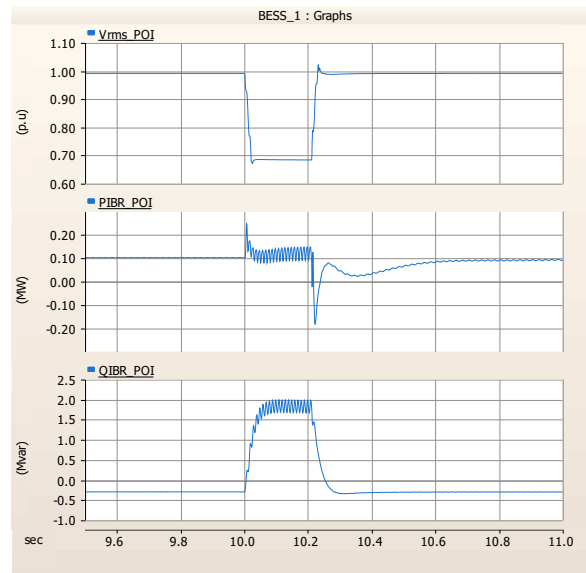


Figure 13. Case 1: RMS voltage, active power and reactive power at the POI

Case 2: Frequency step change

The responses of the hybrid PV-BESS model when subjected to a frequency response test are presented here. At $T = 10$ s, the grid frequency starts to decrease from 50 Hz to 47 Hz in 3 s and remains at 47 Hz until $T = 15$ s when the grid frequency starts ramping up and reach 50 Hz within 3 s.

Figure 14 shows that the hybrid PV-BESS plant responded to the frequency deviation by injecting active power into the grid. The amount of active power generated by the hybrid plant is limited by the 'contract_MW' value specified in the hybrid model. Figure 15 shows the frequency response from the BESS where it transitioned from absorbing around 2 MW of active power to exporting around 0.72 MW during the frequency event.

The active power plot in Figure 16 shows that the PV has very limited respond during the frequency event as the PV is operating in a fixed P and Q mode with the frequency-droop control in the PV inverter disabled ($f_flg = 0$) and with the active power reference signal from the PPC being limited to 0.0.

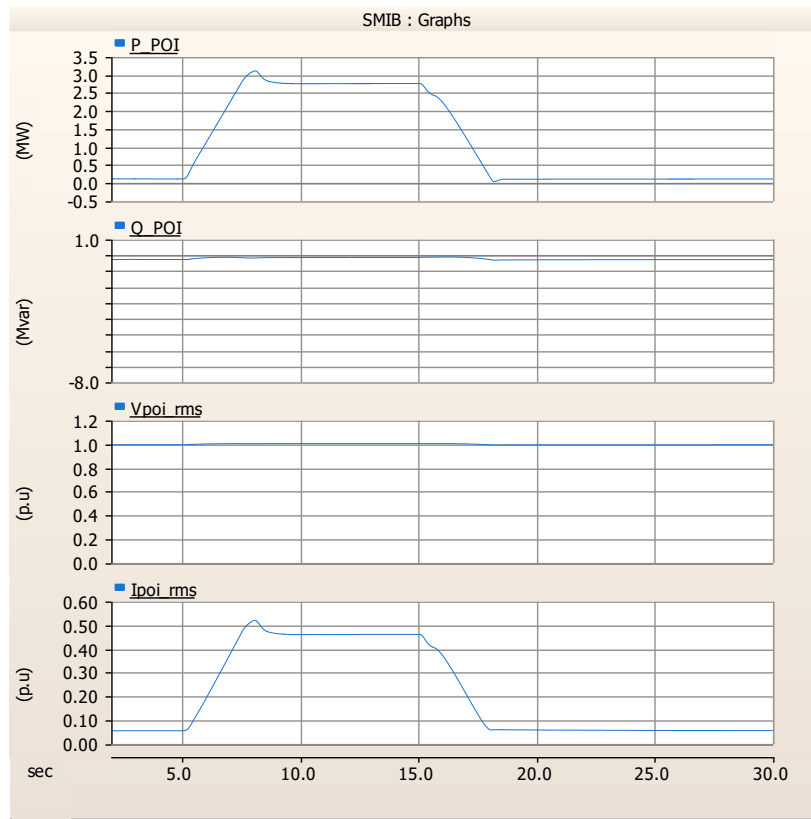


Figure 14. Case 2: Active power, reactive power, voltage and currents at the POI.

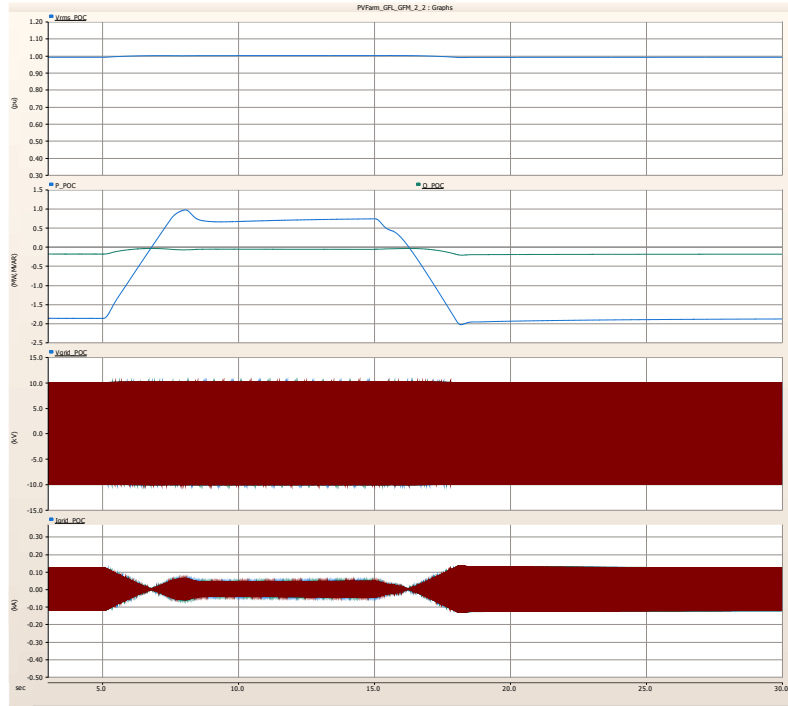


Figure 15. Case 2: Voltages, active power, reactive power and currents at the HV side of the BESS terminal.

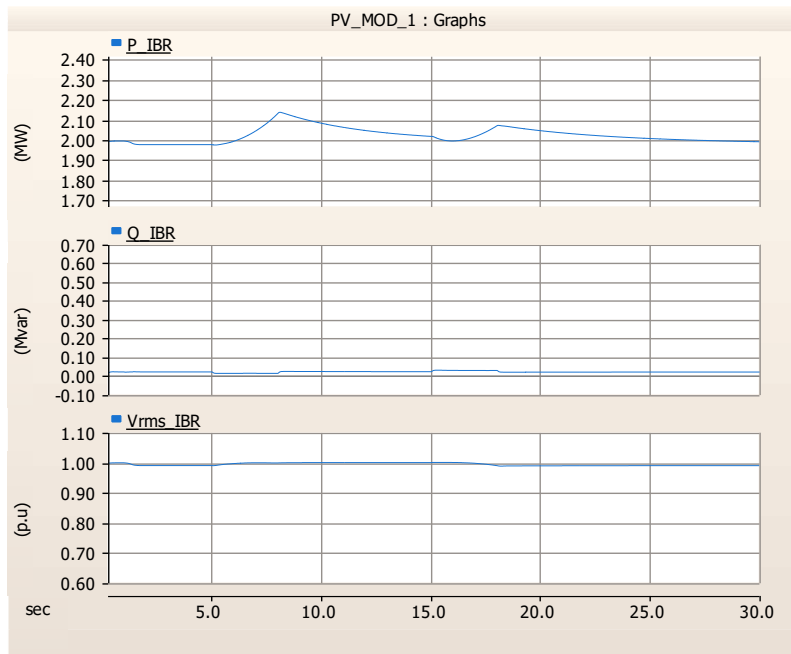


Figure 16. Case 2: Total active power, reactive power and voltage from the PV

Case 3: Voltage step change

The response of the hybrid PV-BESS model when subjected to voltage step changes from the grid are presented here. The first case considered that the BESS is operating as grid following mode and the second case considered that the BESS operating at grid forming mode.

At $T = 10$ s, there is a step increase in the grid voltage from 1 p.u to 1.025 p.u and subsequently at $T = 15$ s, the grid voltage dropped down to 1.0 p.u.

Case 3a: BESS in Grid following mode

The BESS inverter settings as described in Section 2 are considered. Figure 17 shows the hybrid PV-BESS overall plant active power, reactive power and the RMS voltage at the POI. It is noted that with PV operating at fixed P and Q control and the BESS operating at grid following mode, the hybrid plant does not response to the voltage step change. The positive and negative sequence components of the BESS and the PV inverter terminals are shown in Figure 18 and Figure 19.

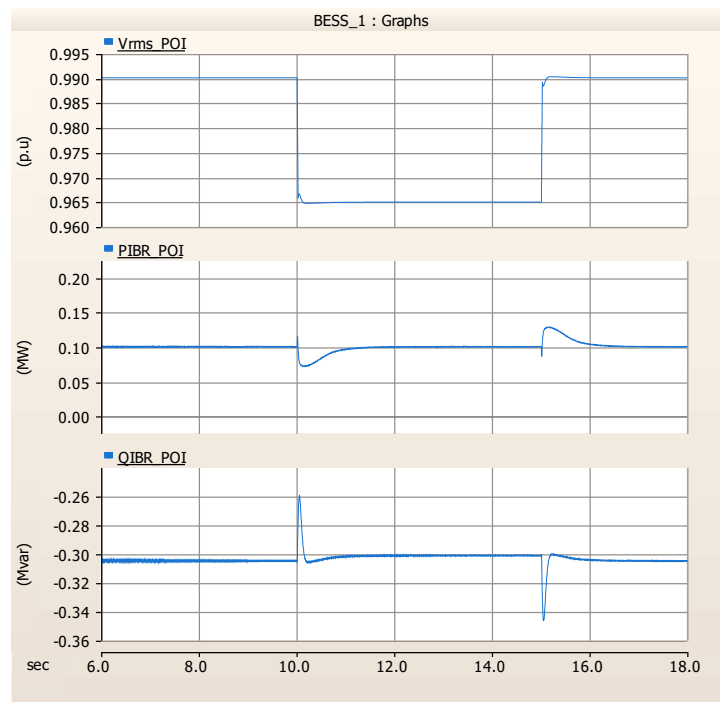


Figure 17. Case 3a: RMS voltage, active power and reactive power at the POI

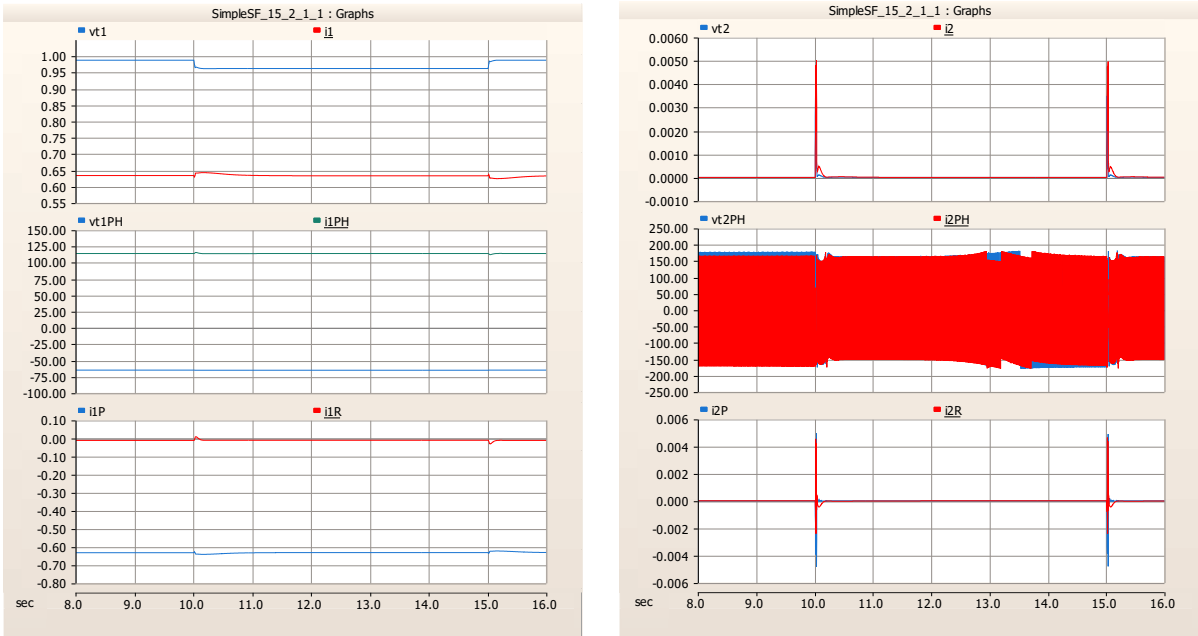


Figure 18. Case 3a: Fundamental frequency components of the BESS inverter terminal voltage and current

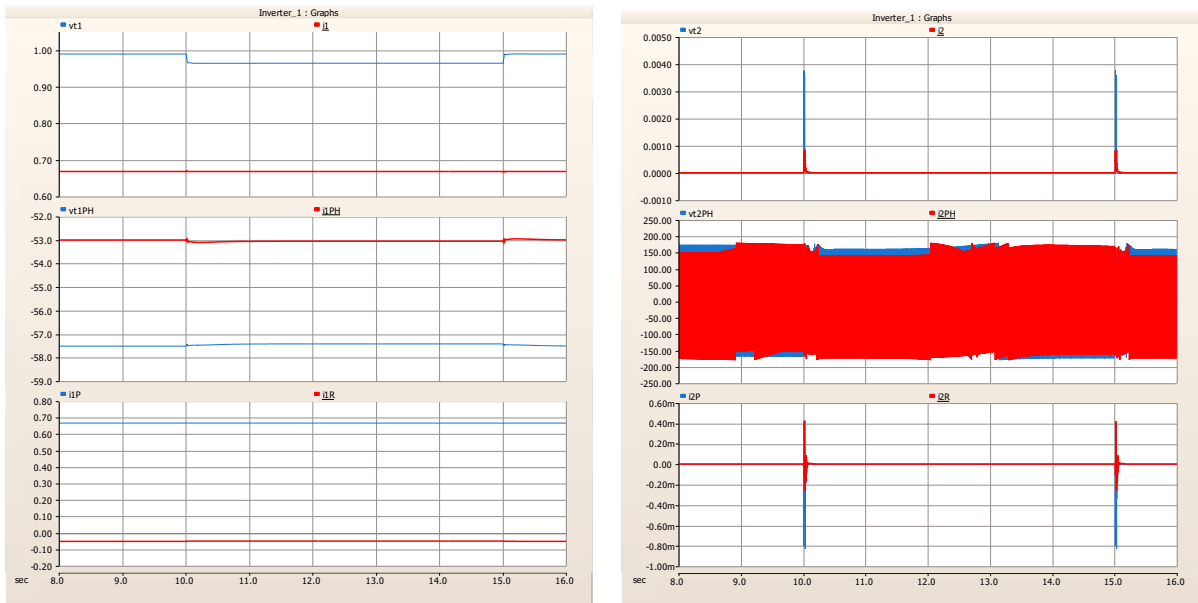


Figure 19. Case 3a: Fundamental frequency components of the PV inverter terminal voltage and current

Case 3b: BESS Grid forming mode

In this section, the voltage step change response of the hybrid PV-BESS model with the BESS operating in PLL-based grid forming mode has been considered. The BESS grid forming mode is achieved with the following inverter model settings:

- wtype = SRF-PLL
- Vflag = 0
- Qflag = 1

Figure 20 shows that the hybrid PV-BESS plant injects reactive power when the voltage at the POI decreases from 1 p.u. to 0.98 p.u. This improves voltage at the POI slightly to 0.99 p.u. during the voltage step event. The positive and negative sequence voltages at the BESS inverter terminal are shown in Figure 21. The results show that the BESS responded to the sudden voltage decrement by injecting positive sequence reactive current.

The positive and negative sequence voltages at the PV inverter terminal are shown in Figure 22, where the positive and negative sequence currents remain constant throughout the voltage step event.

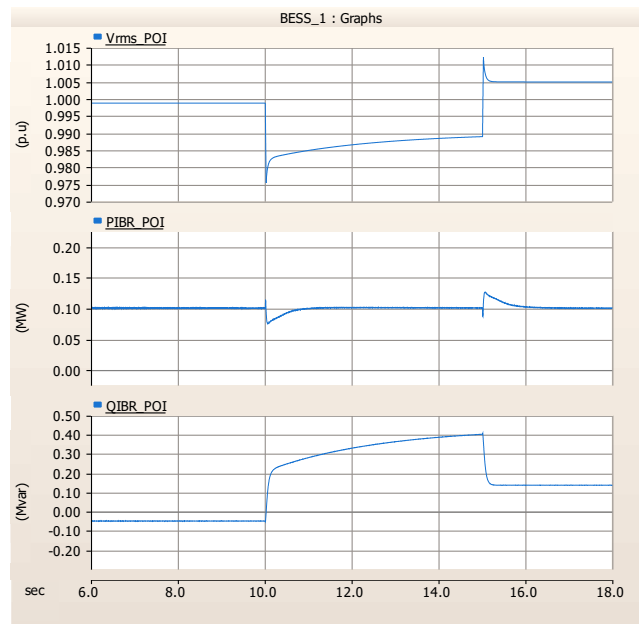


Figure 20. Case 3b: RMS voltage, active power and reactive power at the POI

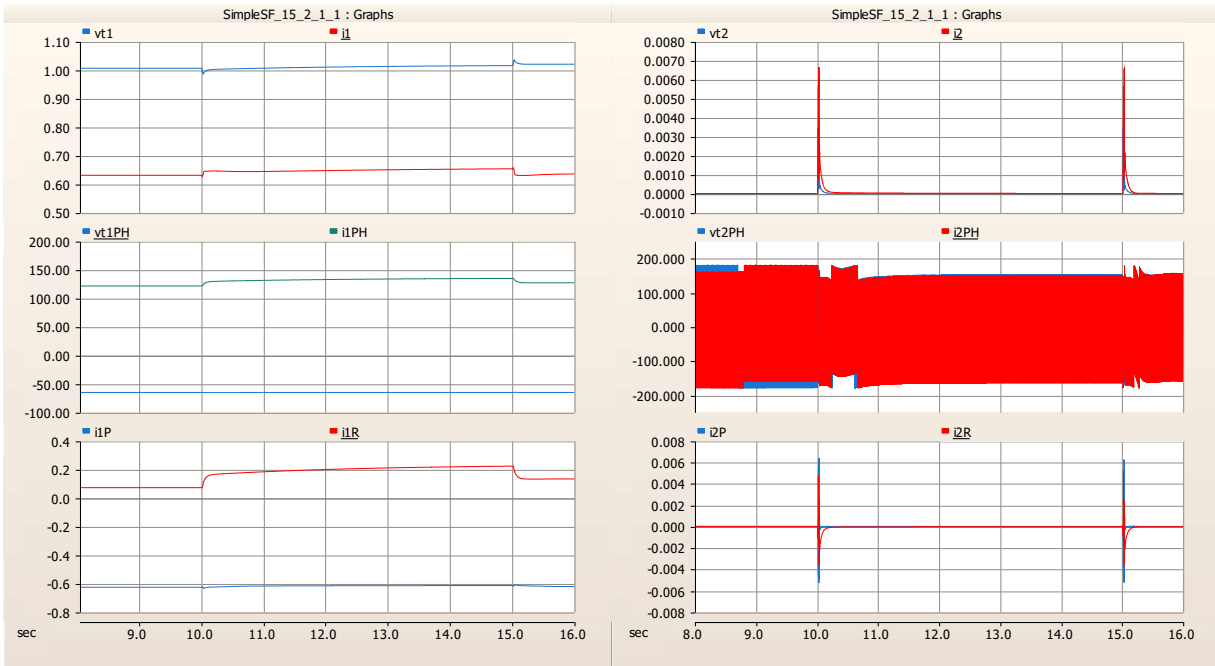


Figure 21. Case 3b: Fundamental frequency components of the BESS inverter terminal voltage and current

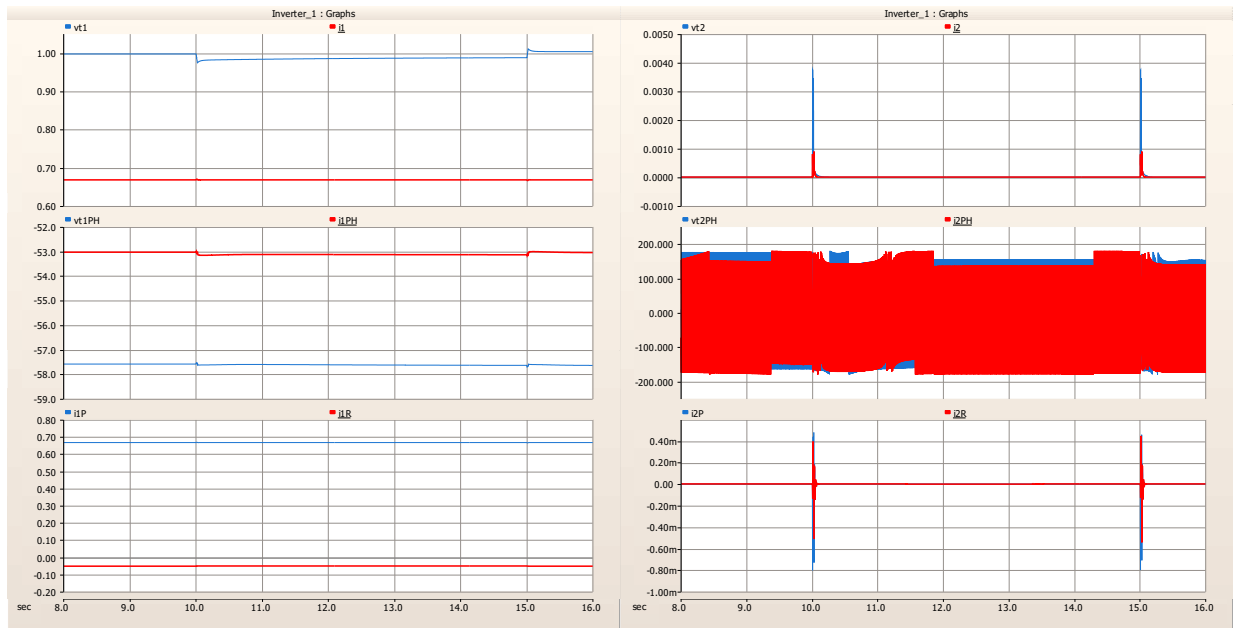


Figure 22. Case 3b: Fundamental frequency components of the PV inverter terminal voltage and current

4 CONCLUSION

The generic hybrid PV-BESS plant EMT model presented in this document can be used for transmission planners for long-term planning studies and to understand the behaviour of a hybrid PV-BESS plant at various network conditions. It also provides flexibility for user to tune the controller settings of the plant controller as well as controller settings at the inverter level to assess their potential impacts of a prospective hybrid PV-BESS plant to the system. This model is not intended to be representative of exact equipment manufacturer controls and neither it is intended to be stable/operational in all system scenarios.

5 REFERENCES

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